SPARK: AGILE TEST INFRASTRUCTURE PAVING THE WAY FOR A SUSTAINABLE EUROPEAN PRESENCE IN SPACE

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The Exploration Company

1.1 European Launch Deficit

324 Global vs. 8 European Launches

Payload Analysis ($N = 316$)

Deficit Trend: 93% (2024) → 90% (2025) → 94% (2026)

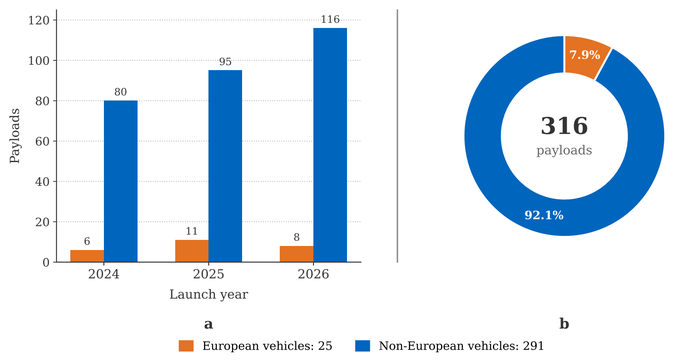


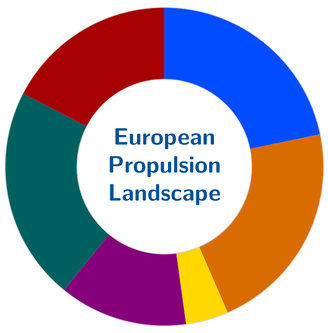
Figure 1: Launcher origin for European payloads (2024–2026).

Bottleneck: Foreign launch reliance demands sovereign testing to develop European vehicles.

1.2 Propulsion Diversity & Modularity Dilemma

Trade-Off: Universal test infrastructure risks high CapEx; bespoke test stands lack versatility.

European Propulsion ($N = 23 \leq 100$ kN)



- **22%** Cryo Methalox
- **22%** Semi-Cryo RP-1
- **22%** Green / In-Space
- **17%** Storable Hypergolics
- **13%** Hybrids
- **4%** Propane

1.3 Co-Development Strategy

1. **Digital Baseline**
   P&ID & 3D CAD Twin outline SPARK roadmap.
2. **Overlap Analysis**
   Identify partner overlap with SPARK roadmap.
3. **Permanent Assets**
   Co-develop shared modules as permanent infrastructure.
4. **Open Interfaces**
   Flanged spools & modular SCADA systems ensure expandability.

2.1 Digital Baseline

KiCad P&ID
- Single source of truth
- Embedded component specs
- Control system tag sync
- Living component catalogues

3D CAD Twin
- Clash detection
- Fast on-site integration
- IP protection through zoning
- Minimal on-site rework

2.2 Modular Control Architecture

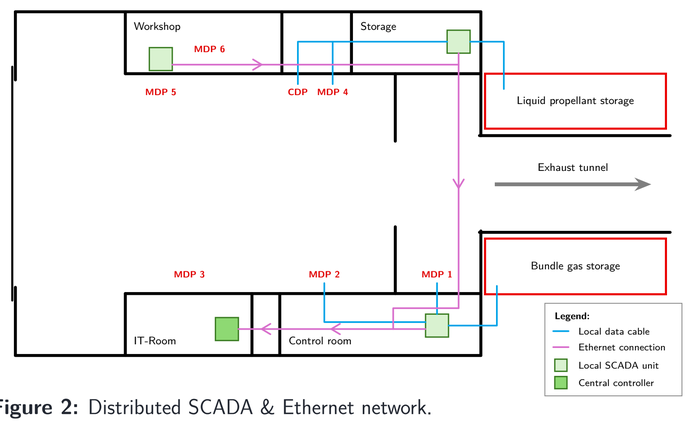


Figure 2: Distributed SCADA & Ethernet network.

SPARK Facility Process Control System (PCS)
Beckhoff IPC · TwinCAT · S.E.A. TestMaster

Closed-Loop Conditioning
- Tank ullage PID regulation
- Automated line chill-down
- Cryogenic sub-cooling

Facility Interlocks
- Facility readiness status
- Gas supply pressure checks
- Master abort triggers

Telemetry & Pressures ↕ Interlocks & Safety Handshake

EtherCAT Bridge Terminal
Galvanic Isolation Boundary · Variable Exchange Interface

Primary EtherCAT Port — *SPARK Facility Network*
Partner EtherCAT Port — *Customer Network*

Setpoints & Process Data ↕ Abort Signals & State Sync

Customer Control Node & Test Sequencer
Partner Automation · Flight Avionics · Customer DAQ

Target Setpoints
- Conditioning target setpoints
- Propellant fill commands
- Line valve actuations

Sequencing & Safety
- Test countdown state
- Engine fire authorization
- Emergency cut-off command

Figure 3: Deterministic EtherCAT gateway architecture.

Pre-Alarm (30% LEL): Automated bridge signal → Orderly shutdown.
Main Alarm (60% LEL): Facility E-stop; UPS emergency safing.

2.3 Co-Engineering: RFA

1. GN_2 Systems
- 300 bar bundles
- 10 bar pneumatic lines
- Bulk evaporator tie-in

2. Fluidics & Ignition
- LOX dump & fill lines
- LOX/RP-1 run tanks
- TEA/TEB supply

3. Safety Architecture
- 2-stage alarm
- Passive safing
- Node UPS backup

4. Control Gateway
- EtherCAT bridge
- Modular architecture
- Cyclic PDO telemetry

3.1 Facility Overview

Site: 500 m^2 hall, 30 m exhaust tunnel.

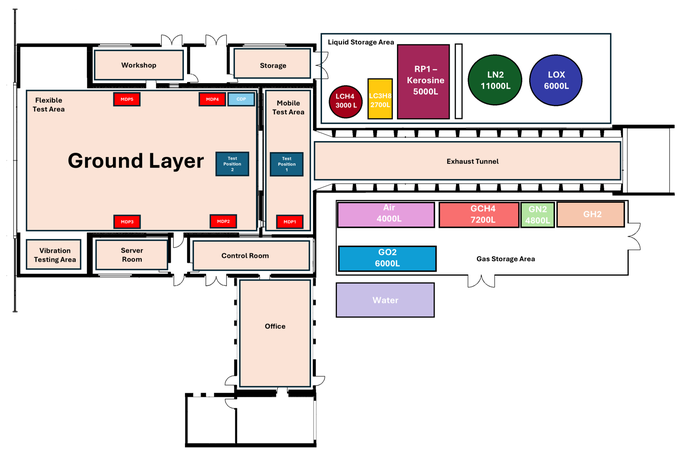


Figure 4: SPARK facility layout.

3.2 SPARK Infrastructure



(a) SPARK Main Entrance



(b) 30 m Exhaust Tunnel



(c) Media Storage Area

3.3 Active Campaigns

OP4/TOSCA: GCH_4/GO_2-LOX/GCH_4 80 bar hot-fire.

TUM ASCENT: Hopper vehicle test bed in MTA.

WARR Rocketry: Integrated vehicle hot-fire tests.

Delta Orbit: Mobile hot-fire operations.

WEP: Transpiration-cooled micro-thruster testing.

Core Transferable Lessons
- Living P&ID
- Deterministic Bridge
- Two-Stage Safing
- Agile Scaling

4. SPARK Operational Test Envelopes & Facility Capabilities

TP1: High Thrust Cell
- **Thrust:** 60 kN max. sea-level
- **Liquids:** LOX, RP-1, LCH_4, TEA/TEB | LC_3H_8
- **Gases:** GN_2, GCO_2, GHe | GO_2, GCH_4
- **Scope:** Subscale / Access to space

TP2: In-Space Bay
- **Thrust:** 0 N–10 kN subscale
- **Liquids:** LOX, H_2O, RP-1, LN_2 | LCH_4, LC_3H_8
- **Gases:** GN_2, GO_2, GCH_4 | GCO_2, GHe, HP-Air
- **Scope:** Subscale / In-space systems

MTA: Mobile Test Area
- **Thrust:** 0–10 kN
- **Liquids:** Dewar-fed
- **Gases:** GN_2, GO_2, GCH_4 | GCO_2, GHe, HP-Air
- **Scope:** Mobile / Self-contained test stands

Media Storage Area
- **Cryo:** 11 kL LN_2, 6 kL LOX, 3 kL LCH_4
- **Fuels:** 3 kL RP-1, 2.7 kL LC_3H_8
- **Gases:** GN_2, GCH_4, GH_2, GO_2, CO_2
- **Run Tanks:** RP-1, LOX | LCH_4, LC_3H_8

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Legend: Operational / Ready | Planned